

Week 6 Report: Model Development

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

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What I did this week

This week I tuned the models with stratified 3-fold cross-validation on the training data. I used PR-AUC as the tuning score because it focuses on how well the rare fraud class is ranked, which is what matters here.

Random Forest clearly leads with a cross-validated PR-AUC of 0.840 and the smallest spread across folds. Logistic Regression (0.746) and Decision Tree (0.738) are close to each other.

Model	Selected settings	CV PR-AUC	SD
Logistic Regression	{C: 1.0}	0.746	0.026
Decision Tree	{max_depth: 20, min_samples_leaf: 10}	0.738	0.017
Random Forest	{max_depth: None, min_samples_leaf: 1}	0.840	0.007

Next week

Compare no resampling, class weights and SMOTE for every model.